

NET512

Designing and Implementing a Secure, Scalable, and Automated Enterprise Network

Weekly Practical Plan — NQF Level 5 | 17 Credits

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Week 2: Cisco Device Functions (5 Marks)

Objective: Explore how switches learn MAC addresses and how routers build and use routing tables — the two core lookups that get a frame/packet to the right place.

ROUTER	IP ADDRESS	SUBNET MASK
1	10.0.12.1	255.255.255.252
1	10.0.134.1	255.255.255.0
2	10.0.12.2	255.255.255.252
3	10.0.134.3	255.255.255.0
4	10.0.134.4	255.255.255.0

5. Verify the switch MAC address table with 'show mac address-table

```
SW1>enable
SW1#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.173b.c901    DYNAMIC Fa0/1
1       00e0.a3c8.c718    DYNAMIC Fa0/24
10      00d0.ffc1.bb01    DYNAMIC Fa0/2
10      00e0.a3c8.c718    DYNAMIC Fa0/24
20      0002.173b.c902    DYNAMIC Fa0/3
20      00e0.a3c8.c718    DYNAMIC Fa0/24
SW1#
```

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```
SW2>enable
SW2#show mac addresss-table
^
% Invalid input detected at '^' marker.

SW2#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.428b.6018    DYNAMIC Fa0/24
20      0090.2b96.b801    DYNAMIC Fa0/4
20      00e0.f973.0e02    DYNAMIC Fa0/3
SW2#
```

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6. Log into the routers and note down the MAC address of each interface

```
R1#show interface g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0002.173b.c901 (bia 0002.173b.c901)
Internet address is 10.0.12.1/30
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
--More--
```

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```
R2>enable
R2#show interface g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00d0.ffc1.bb01 (bia 00d0.ffc1.bb01)
  Internet address is 10.0.12.2/30
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
--More--
```

```
R3>enable
R3#show interface g0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00e0.f973.0e02 (bia 00e0.f973.0e02)
  Internet address is 10.0.134.3/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
--More--
```

```
R4>enable
R4#show interface g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 0090.2b96.b801 (bia 0090.2b96.b801)
  Internet address is 10.0.134.4/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
--More--
```

7. Verify connectivity between the routers using PING

```

R1>enable
R1#ping 10.0.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.12.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#ping 10.0.134.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.134.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/12 ms

R1#ping 10.0.134.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.134.4, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/11 ms

R1#

```

8. View the dynamically learned MAC addresses on the switch.

```

SW1>enable
SW1#show mac address-table dynamic
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1      0002.173b.c901     DYNAMIC   Fa0/1
1      00e0.a3c8.c718     DYNAMIC   Fa0/24
10     00d0.ffc1.bb01     DYNAMIC   Fa0/2
10     00e0.a3c8.c718     DYNAMIC   Fa0/24
20     0002.173b.c902     DYNAMIC   Fa0/3
20     0090.2b96.b801     DYNAMIC   Fa0/24
20     00e0.a3c8.c718     DYNAMIC   Fa0/24
20     00e0.f973.0e02     DYNAMIC   Fa0/24
SW1#

```

```

SW2>enable
SW2#show mac address-table dynamic
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1      0001.428b.6018     DYNAMIC   Fa0/24
1      0002.173b.c901     DYNAMIC   Fa0/24
20     0002.173b.c902     DYNAMIC   Fa0/24
20     0090.2b96.b801     DYNAMIC   Fa0/4
20     00e0.f973.0e02     DYNAMIC   Fa0/3
SW2#

```

9. Clear the MAC address table on the switch, then check it again.

```

SW1#clear mac address-table dynamic
SW1#show mac address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1      00e0.a3c8.c718     DYNAMIC   Fa0/24
10     00d0.ffc1.bb01     DYNAMIC   Fa0/2
10     00e0.a3c8.c718     DYNAMIC   Fa0/24
20     00e0.a3c8.c718     DYNAMIC   Fa0/24
SW1#

```



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10. Examine the routing table on the routers with 'show IP route'

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/0
L       10.0.12.1/32 is directly connected, GigabitEthernet0/0
C       10.0.134.0/24 is directly connected, GigabitEthernet0/1
L       10.0.134.1/32 is directly connected, GigabitEthernet0/1

R1#

```

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```

R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/0
L       10.0.12.2/32 is directly connected, GigabitEthernet0/0

R2#

```

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```

R3>enable
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.134.0/24 is directly connected, GigabitEthernet0/1
L       10.0.134.3/32 is directly connected, GigabitEthernet0/1

R3#

```



```

R4>enable
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.134.0/24 is directly connected, GigabitEthernet0/0
L       10.0.134.4/32 is directly connected, GigabitEthernet0/0

R4#

```

11. Configure IP addresses on the router interfaces as per the topology.

```

R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.12.1      YES manual up          up
GigabitEthernet0/1  10.0.134.1     YES manual up          up
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down

R1#

```

```
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.12.2      YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#
```

```
R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  administratively down down
GigabitEthernet0/1  10.0.134.3     YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R3#
```

```
R4#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.134.4     YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R4#
```

12. Confirm the new connected routes appear in the routing table

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.134.0/24 is directly connected, GigabitEthernet0/1
L    10.0.134.1/32 is directly connected, GigabitEthernet0/1
R1#
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.2/32 is directly connected, GigabitEthernet0/0
R2#
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.134.0/24 is directly connected, GigabitEthernet0/1
L    10.0.134.3/32 is directly connected, GigabitEthernet0/1
R3#
```

```

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.134.0/24 is directly connected, GigabitEthernet0/0
L       10.0.134.4/32 is directly connected, GigabitEthernet0/0

R4#s

```



13. Configure static routes between the routers.

```

R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.134.0 255.255.255.0 10.0.12.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#

```



```

R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.12.0 255.255.255.252 10.0.134.1
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#

```

```

R4#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route 10.0.12.1 255.255.255.252 10.2.134.1
%Inconsistent address and mask
R4(config)#ip route 10.0.12.0 255.255.255.252 10.0.134.1
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console

R4#copy running-config startuo-config
^
% Invalid input detected at '^' marker.

R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#

```

14. Confirm the new static routes appear in the routing table

```

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/0
L       10.0.12.2/32 is directly connected, GigabitEthernet0/0
S       10.0.134.0/24 [1/0] via 10.0.12.1

R2#

```

```

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.12.0/30 [1/0] via 10.0.134.1
C       10.0.134.0/24 is directly connected, GigabitEthernet0/1
L       10.0.134.3/32 is directly connected, GigabitEthernet0/1

R3#

```

```

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.0.12.0/30 [1/0] via 10.0.134.1
C       10.0.134.0/24 is directly connected, GigabitEthernet0/0
L       10.0.134.4/32 is directly connected, GigabitEthernet0/0

R4#

```



15. After clearing the MAC address table, do you see any MAC addresses reappear? Why or why not?

Yes. MAC addresses reappeared because the switch dynamically relearns them when it receives traffic from connected devices. Clearing the MAC address table removes the learned entries, but they are automatically learned again when devices communicate.